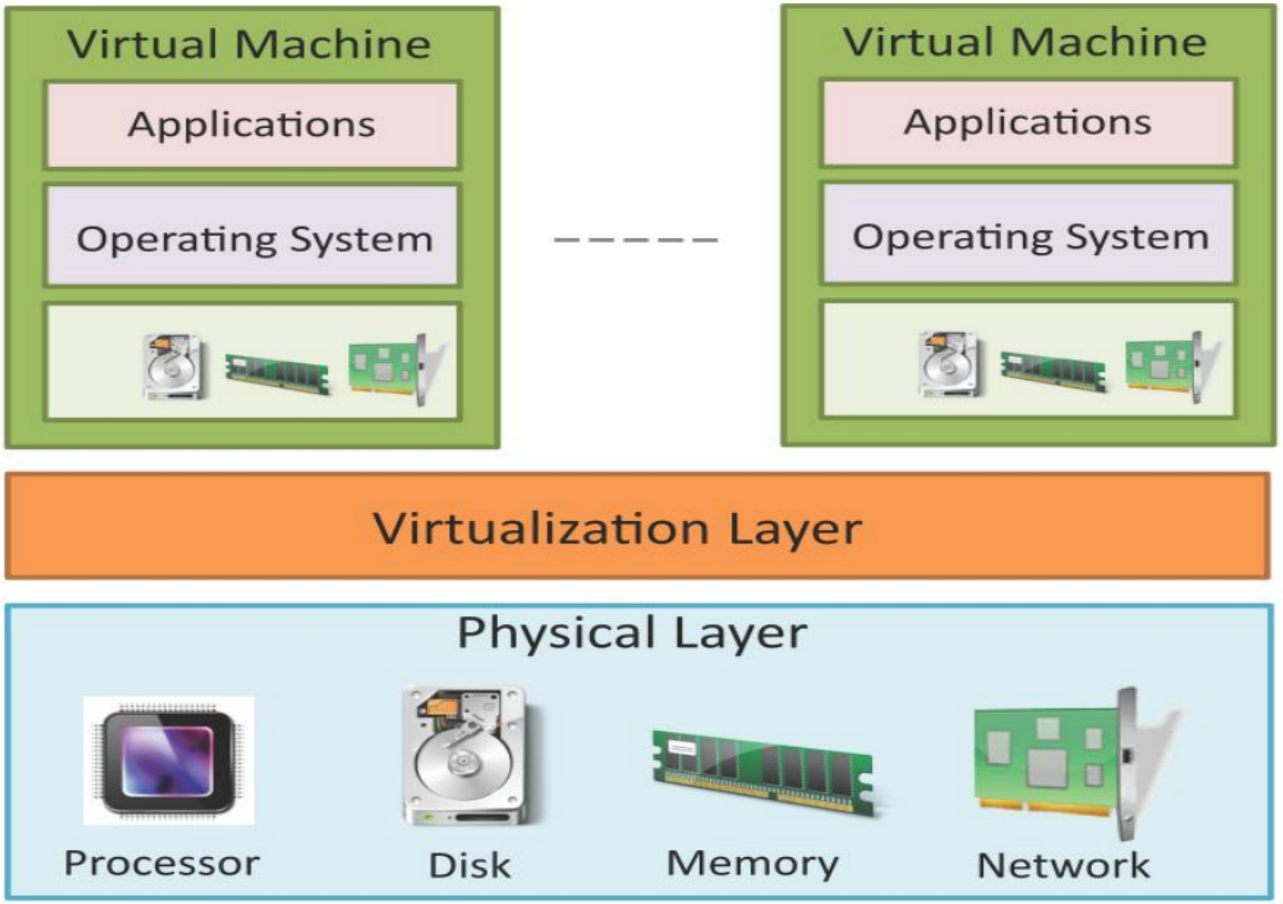
IT Workshop II (Cloud Computing): Cloud Concepts and Technologies

August 4, 2026

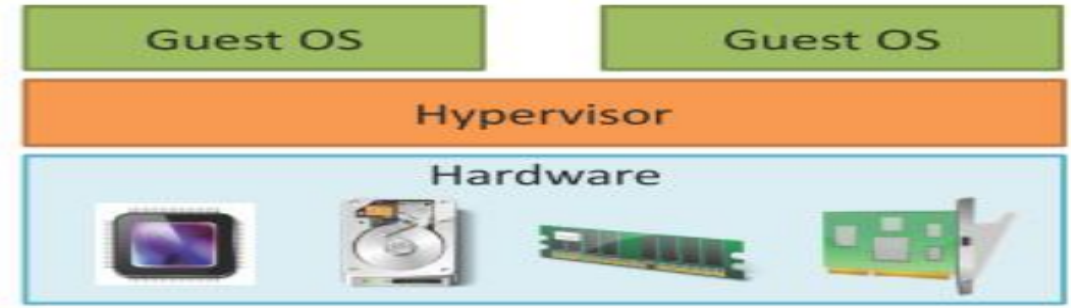
Virtualization



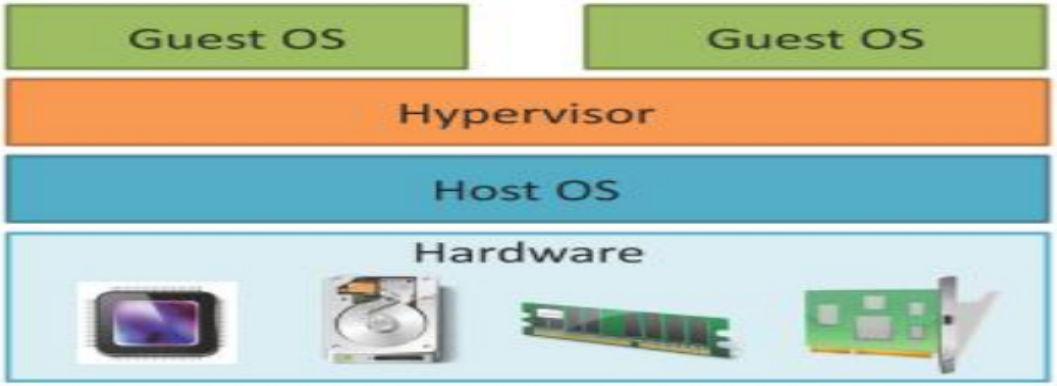
Hypervisor

- The virtualization layer consists of a hypervisor or a virtual machine monitor (VMM).
- Hypervisor presents a virtual operating platform to a guest operating system (OS).
- **Type-1 Hypervisor:** Type-I or the native hypervisors run directly on the host hardware and control the hardware and monitor the guest operating systems.
- **Type-2 Hypervisor:** Type 2 hypervisors or hosted hypervisors run on top of a conventional (main/host) operating system and monitor the guest operating systems.

Hypervisor



Type-1 Hypervisor



Type-2 Hypervisor

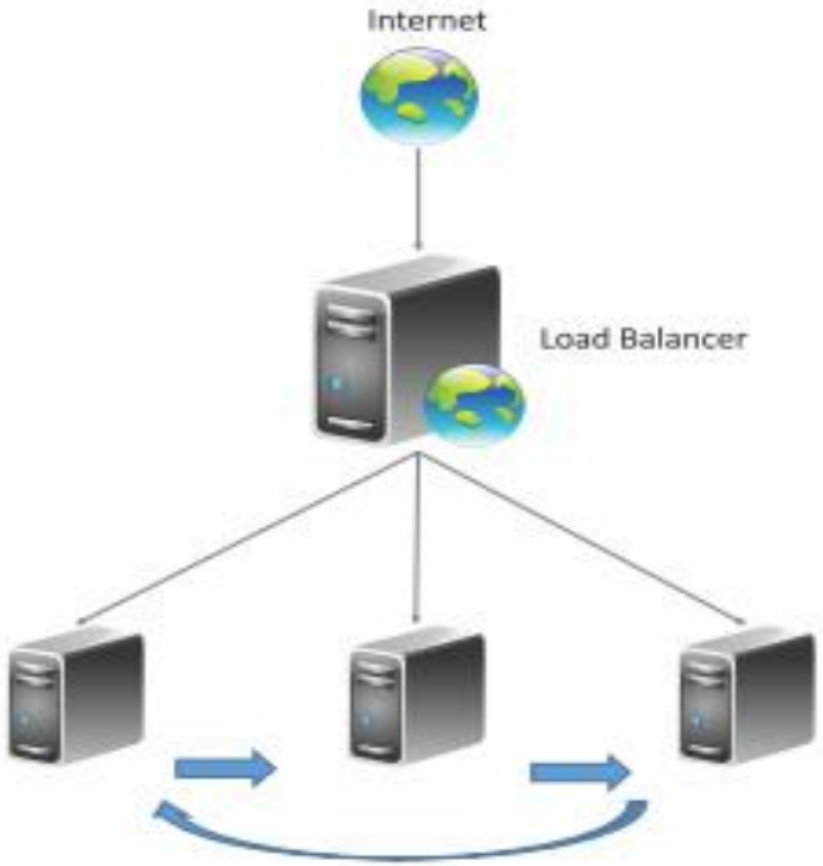
Types of Virtualization

- **Full Virtualization**: The virtualization layer completely decouples the guest OS from the underlying hardware. The guest OS requires no modification and is not aware that it is being virtualized.
- **Para-Virtualization**: The guest OS is modified to enable communication with the hypervisor to improve performance and efficiency. The guest OS kernel is modified to replace non-virtualizable instructions with hyper-calls that communicate directly with the virtualization layer hypervisor.
- **Hardware Virtualization**: In hardware assisted virtualization, privileged and sensitive calls are set to automatically trap to the hypervisor.

Load Balancing

- Cloud computing resources can be scaled up on demand to meet the performance requirements of applications.
- Load balancing distributes workload accross multiple servers to meet the application workloads.
- The goals of load balancing techniques are to achieve maximum utilization of resources, minimizing the response times, maximizing throughput.
- With load balancing, cloud-based applications can achieve high availability and reliability.

Load Balancing



Scalability and Elasticity

- Multi-tier applications such as e-commerce, social networking, business-to-business, etc. can experience rapid changes in their traffic.
- Modern web applications have multiple tiers of deployment with varying number of servers in each tier.
- When the workloads of applications increase, the traditional approaches have been either to scale up or scale out.
- **Scale Up:** Updating the hardware resources (adding additional computing, memory, storage or network resources).
  Scale Out: Addition of more resources of the same type.

The End